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# Kelvin Probe Force Microscopy and Raman Spectroscopy of Graphene Grown on 1095 Steel

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## Introduction

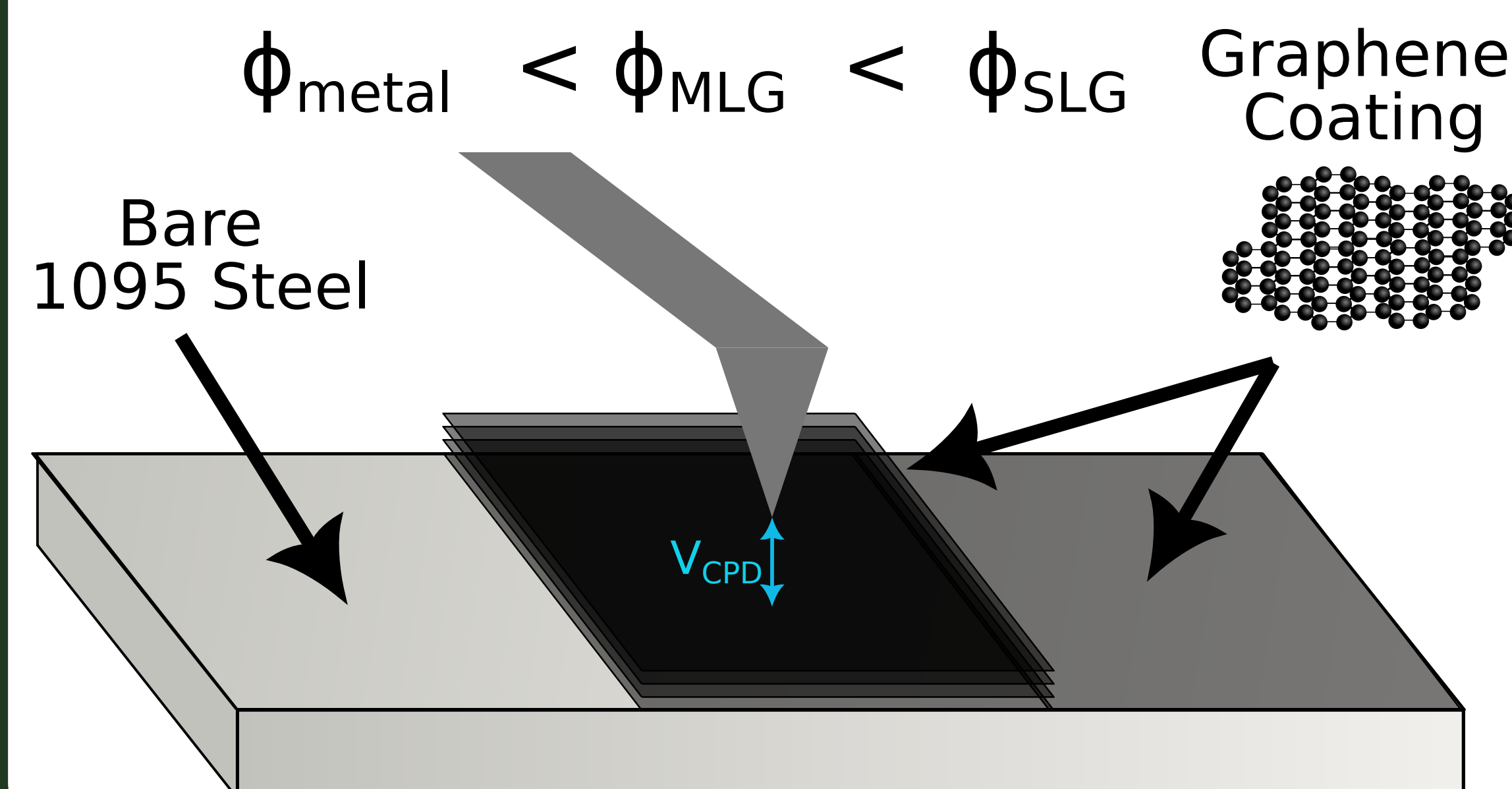
Kelvin probe force microscopy (KPFM) is an atomic force microscope (AFM) technique well suited for evaluating graphene as an anticorrosive coating for steel, because of the close relationship between electronic work function ( $\phi$ ) and electrochemical corrosion potential ( $E_{\text{corr}}$ ) [1]. Graphene's chemical inertness, impermeability, and electronic properties make it a promising corrosion barrier, with work function influenced by layer number, defect density and graphene - substrate interactions.

Here, graphene was grown directly with chemical vapor deposition (CVD) on 1095 steel using acetylene and hydrogen forming gases at 600 °C. The experimental design was based on previous results showing how CVD conditions, alloy composition, and microstructures influence graphene layer number and defect density on steel substrates [2]. Characterization of coatings used Raman spectroscopy, tapping mode AFM/KPFM, and electrochemical testing in simulated seawater.

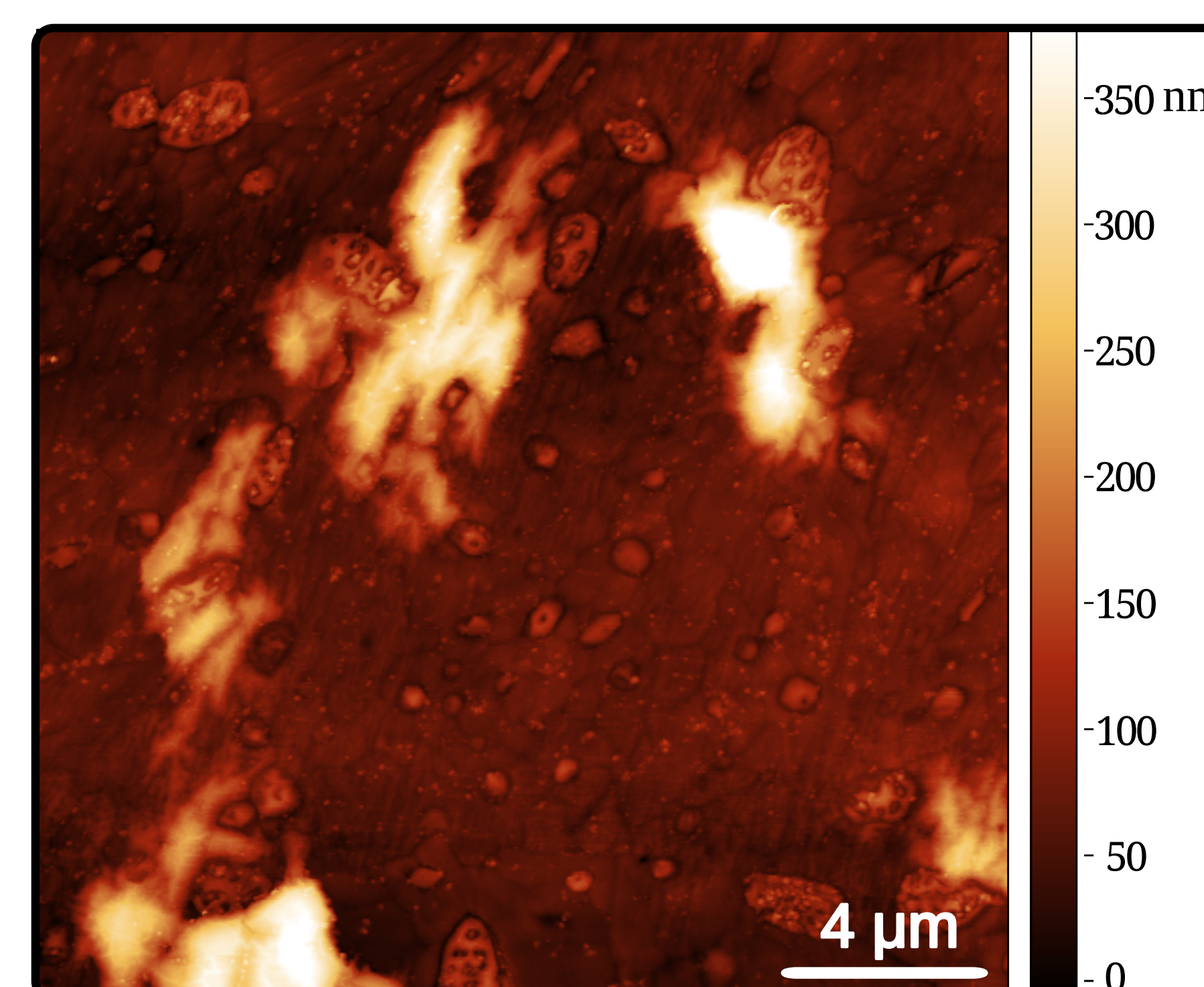
## Theory

In KPFM, a compensating bias, referred to as the contact potential difference ( $V_{\text{CPD}}$ ), is applied between a metal-coated probe and the sample to nullify electrostatic forces. This allows spatial mapping of relative work functions across the surface [3].

- Uncoated steel has a relatively low  $\phi$ , resulting in a high  $E_{\text{corr}}$ .
- Single and few layer graphene (SLG, FLG) grown directly on steel strongly chemisorb to the substrate, resulting in a sharp increase to  $\phi$ .
- Additional layers shield this effect, causing multilayer graphene (MLG) to have a  $\phi$  between bare steel and SLG.

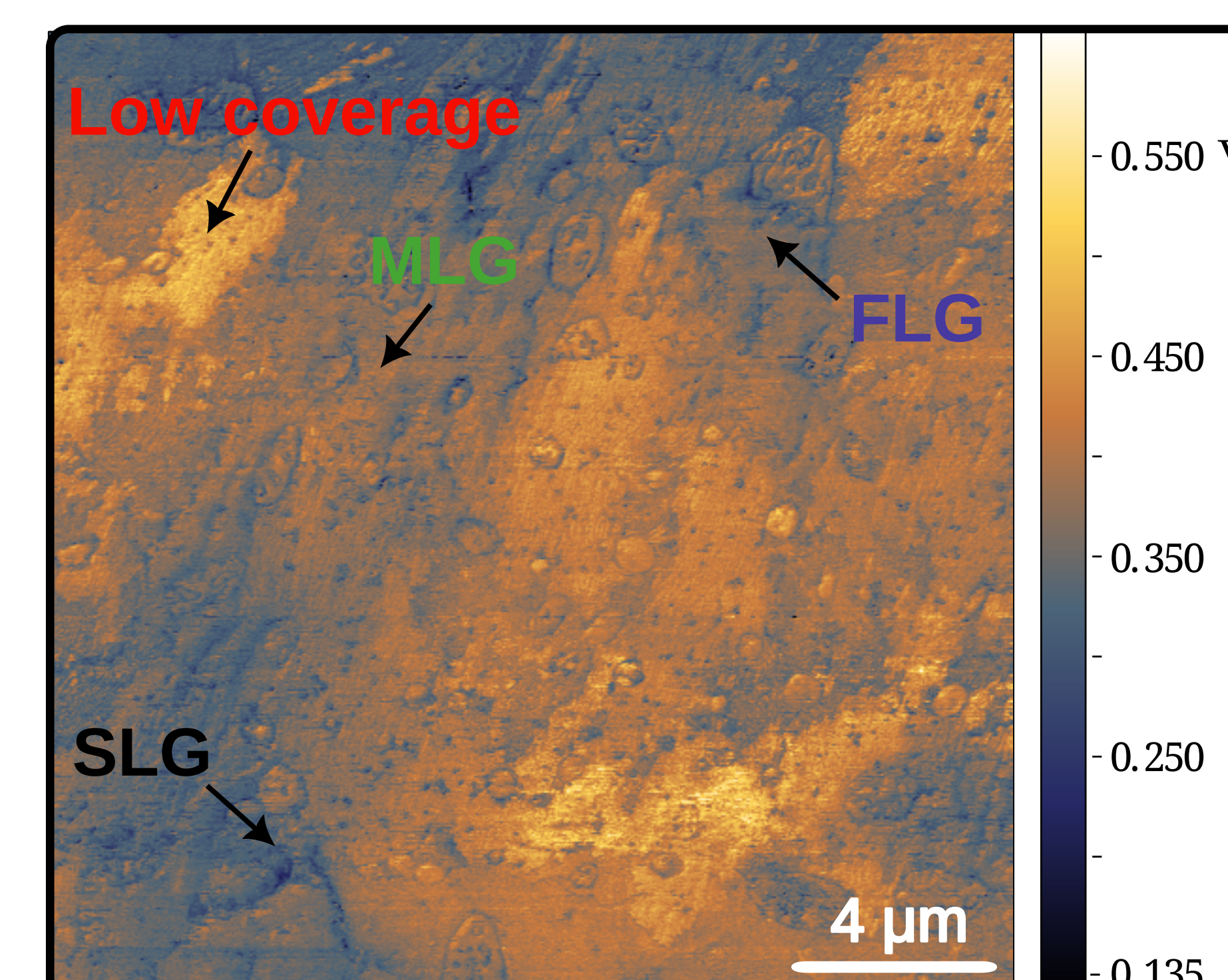


## AFM and KPFM



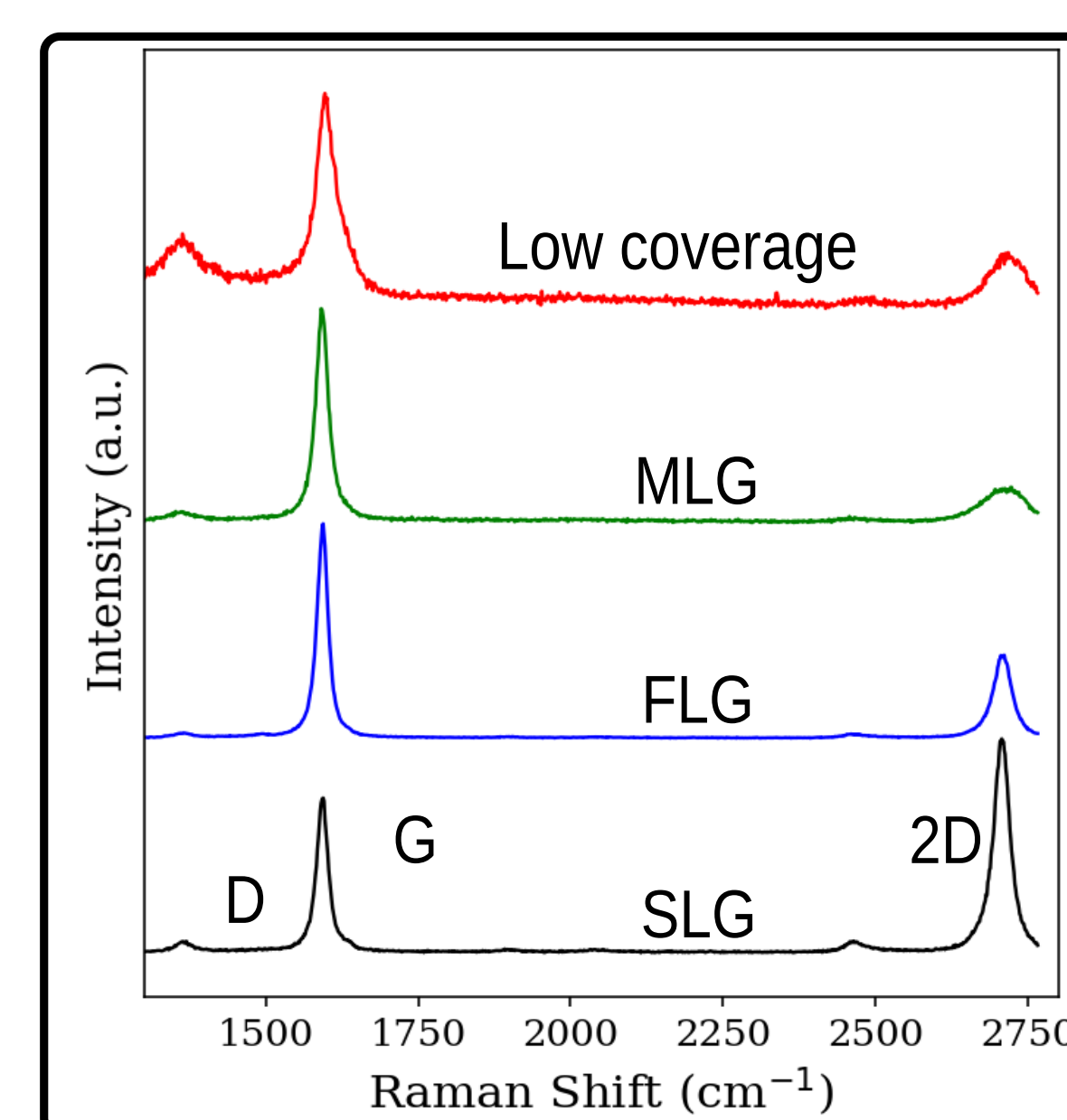
- AFM topography images are limited by surface structure scale compared to the thickness and size of graphene domains.

- With KPFM, contrast is decoupled from topography, relying on  $\phi$  relative to the AFM probe to enable characterization of graphene areas on the rough substrate.



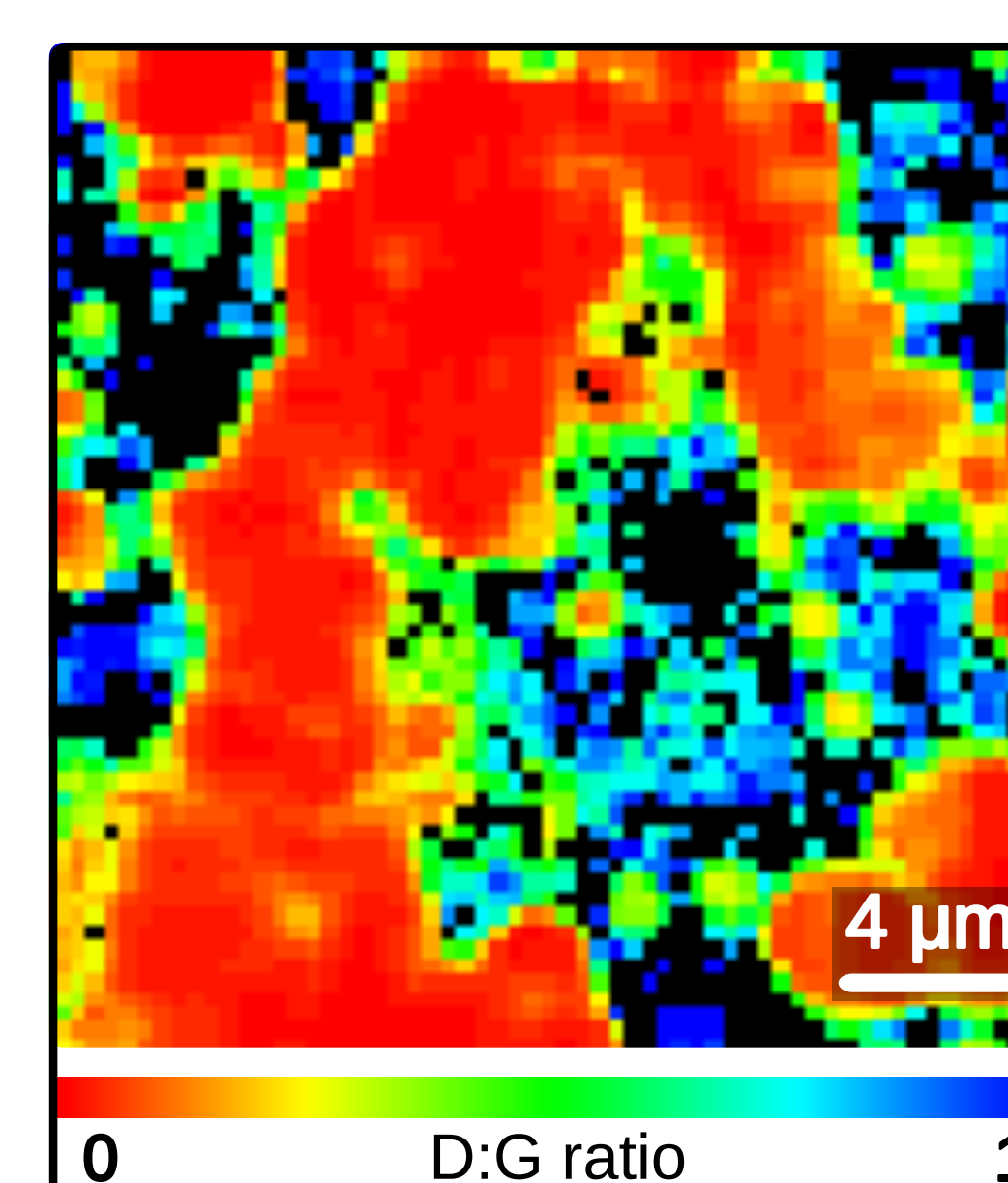
- In the KPFM image, strongly chemisorbed SLG and FLG have the lowest  $V_{\text{CPD}}$ , indicating a higher relative  $\phi$ , while MLG has intermediate  $V_{\text{CPD}}$ .
- Bare steel and areas with low coverage have the highest  $V_{\text{CPD}}$ . These contrast differences show the value of KPFM characterization of graphene coated steel.

## Raman Spectroscopy



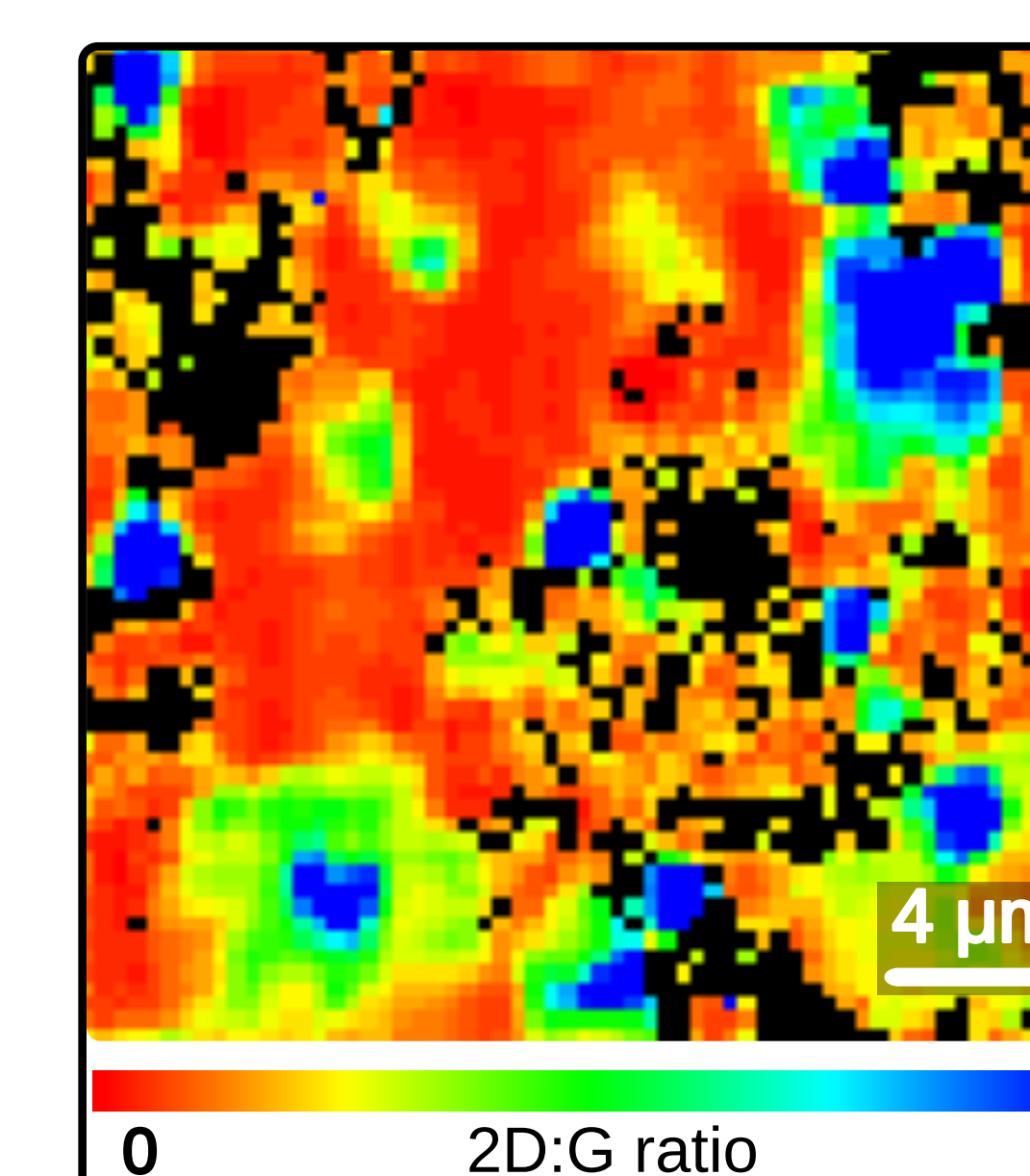
Raman point spectra corresponding to labels in the KPFM image verify:

- Low coverage areas have high  $V_{\text{CPD}}$  (red spectrum with high D:G and low 2D:G peak ratios)
- Low  $V_{\text{CPD}}$  areas are MLG (green), FLG (blue) and SLG (black).



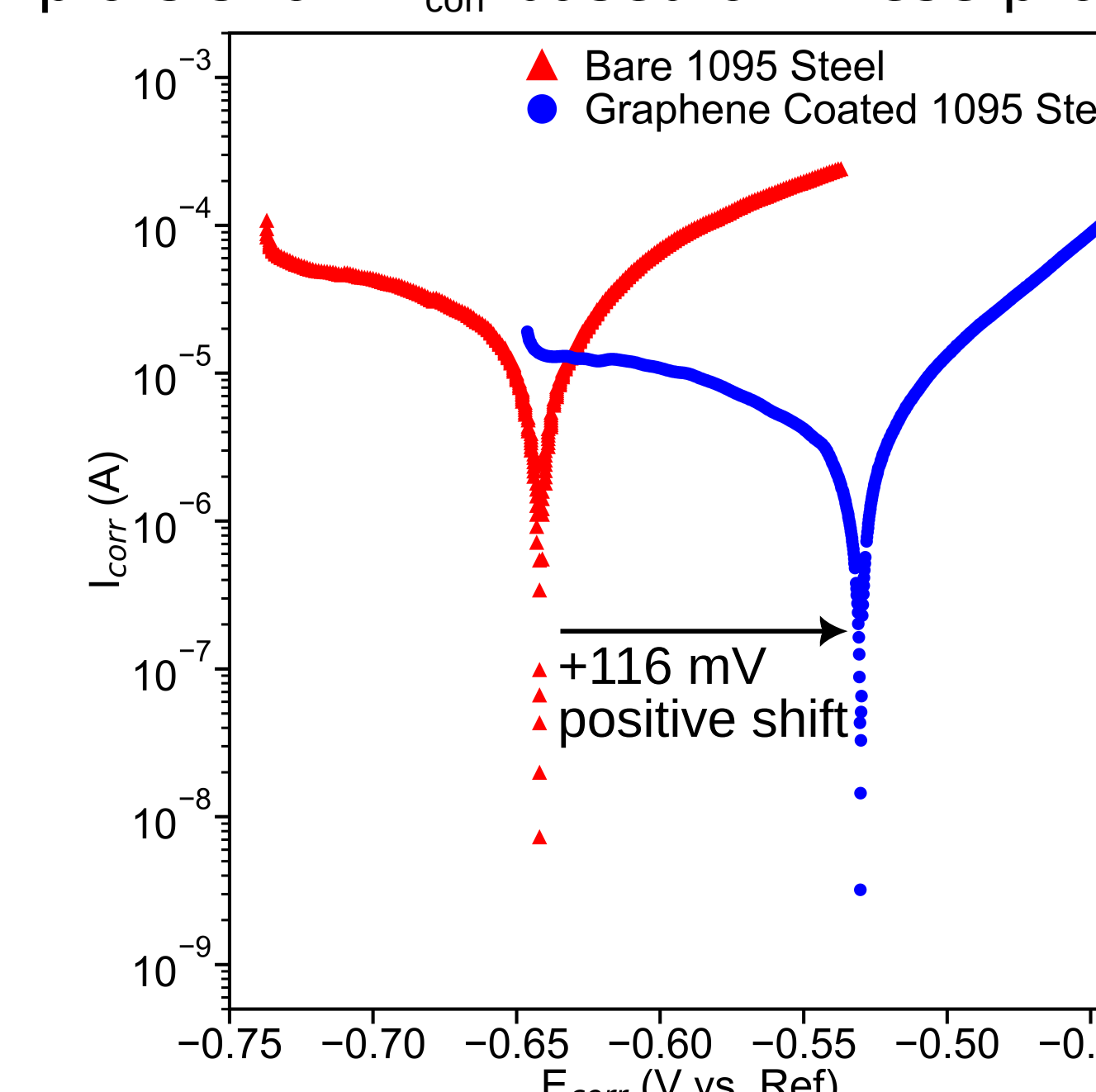
Raman mapping shows a strong spatial correlation between high coverage, thin graphene and low  $V_{\text{CPD}}$ .

- The map of G peak position shows FLG and SLG areas are blue-shifted, supporting the chemisorption-induced doping hypothesis [4]
- The 2D:G peak ratio map shows lowest  $V_{\text{CPD}}$  arises from areas with the highest 2D:G ratio.



## Corrosion Testing

The electronic work function of steel correlates with its ionic work function - a kinetically controlled potential derived from the corrosion processes on the steel surface [5]. Tafel plots show  $E_{\text{corr}}$  based on these processes.



High coverage SLG/FLG coatings on 1095 exhibited a 116 mV positive shift in corrosion potential, validating the role of graphene coatings for tailored steel surface potential.

## Conclusion

KPFM effectively reveals graphene grown on steel, and the importance of layer number and coverage on work function. While high-coverage, low-defect MLG increases the work function, stronger interactions between FLG or SLG and the steel surface raise the work function even more. Combined with Raman spectroscopy, KPFM helps identify suitable CVD parameters that promote high coverage, low defect density, and controlled layer numbers for graphene anticorrosive coatings on steel.

## References

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